AES(Advanced Encryption Standard)

Overview

- Introduction of the AES algorithm
- Internal structure of AES
  - Byte Substitution layer
  - Diffusion layer
  - Key Addition layer
  - Key schedule
- Decryption
- Issues

Introduction

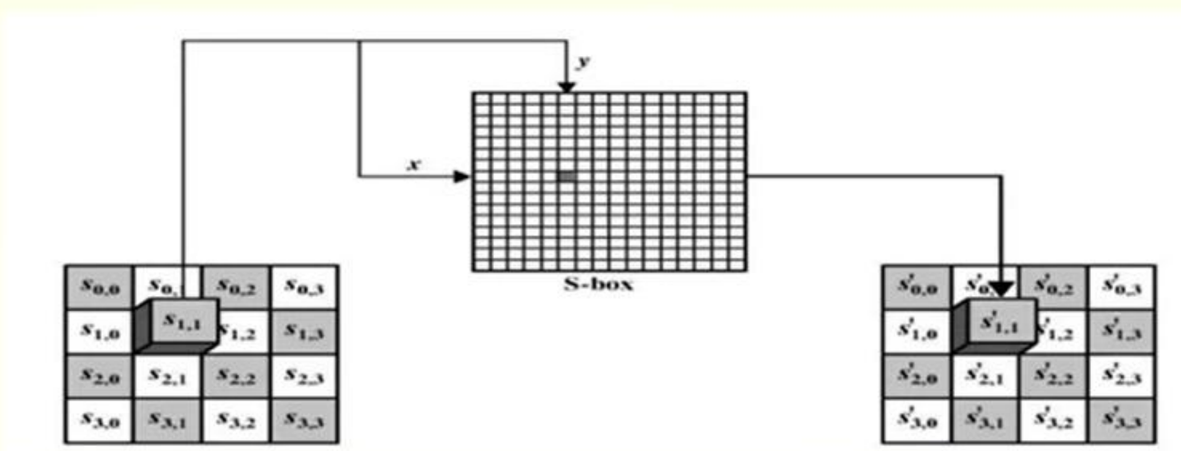
- AES is the most widely used symmetric cipher today
- The algorithm for AES was chosen by the US *National Institute of Standards and Technology* (NIST) in a multi-year selection process
- The requirements for all AES candidate submissions were:
  - Block cipher with **128-bit block size**
  - **Three supported key lengths**: 128, 192 and 256 bit
  - Security relative to other submitted algorithms
  - **Efficiency** in software and hardware

Chronology

- The need for a new block cipher announced by NIST in January, 1997
- 15 candidates algorithms accepted in August, 1998
- 5 finalists announced in August, 1999:
  - *Mars* – IBM Corporation
  - *RC6* – *RSA* Laboratories
  - *Rijndael* – J. Daemen & V. Rijmen
  - *Serpent* – Eli Biham et al.
  - *Twofish* – B. Schneier et al.
- In October 2000, *Rijndael* was chosen as the AES
- AES was formally approved as a US federal standard in November 2001

High-level description of r-round AES

1. Given a plaintext X, initialize **state** to be X and perform an operation **Add round key**, which x-ors the round key with **state**.
2. For each of the first $r-1$ rounds, perform a substitution operation called **SubBytes** on **state** using an S-box; perform a permutation **ShiftRows** on **state**; perform an operation **MixColumns** on **state**; and perform **AddRoundKey**.
3. Perform **SubBytes**; perform **ShiftRows**; and perform **AddRoundKey**.
4. Define the ciphertext Y to be **state**.



Substitute Bytes Stage of the AES algorithm.

Substitute Bytes

- This stage (known as SubBytes) is simply a table lookup using a 16×16 matrix of byte values called an **s-box**.
- This matrix consists of all the possible combinations of an 8 bit sequence ($2^8 = 16 \times 16 = 256$).
- However, the s-box is not just a random permutation of these values and there is a well defined method for creating the s-box tables.
- The designers of Rijndael showed how this was done unlike the s-boxes in DES for which no rationale was given.

		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
x	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

(a) S-box

Algebraic formulation of AES S-box

- In contrast to the S-boxes in DES, which are apparently "random" substitution, the AES S-box can be defined algebraically.
- AES S-box involves operations in the finite field:

$$F_{2^8} = Z_2[x]/(x^8 + x^4 + x^3 + x + 1).$$

- Let FieldInv denote the multiplicative inverse of a field element.
- Let BinaryToField convert a byte to a field element; and FieldToBinary perform the inverse conversion.

Algorithm SubBytes($a_7a_6a_5a_4a_3a_2a_1a_0$)
external: FieldInv, BinaryToField, FieldToBinary
    $z \leftarrow$ BinaryToField($a_7a_6a_5a_4a_3a_2a_1a_0$)
    if $z \neq 0$ **then** $z \leftarrow$ FieldInv(z)
    $(a_7a_6a_5a_4a_3a_2a_1a_0) \leftarrow$ FieldToBinary(z)
    $(c_7c_6c_5c_4c_3c_2c_1c_0) \leftarrow (01100011)$
    comment: all subscripts are to be reduced modulo 8
    for $i \leftarrow 0$ **to** 7
        do $b_i \leftarrow (a_i + a_{i+4} + a_{i+5} + a_{i+6} + a_{i+7} + c_i) \bmod 2$
    return $(b_7b_6b_5b_4b_3b_2b_1b_0)$

Example

- Suppose we begin with (hexadecimal) $\{53\}$, which is 01010011 in binary.
- The corresponding field element is:

$$x^6 + x^4 + x + 1.$$

- The multiplicative inverse (in F_{2^8}) can be shown to be

$$x^7 + x^6 + x^3 + x.$$

Therefore, in binary notation, we have

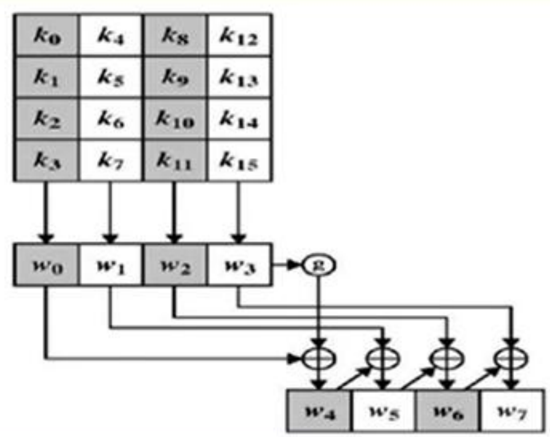
$$(a_7 a_6 a_5 a_4 a_3 a_2 a_1 a_0) = (11001010).$$

- Next, we compute

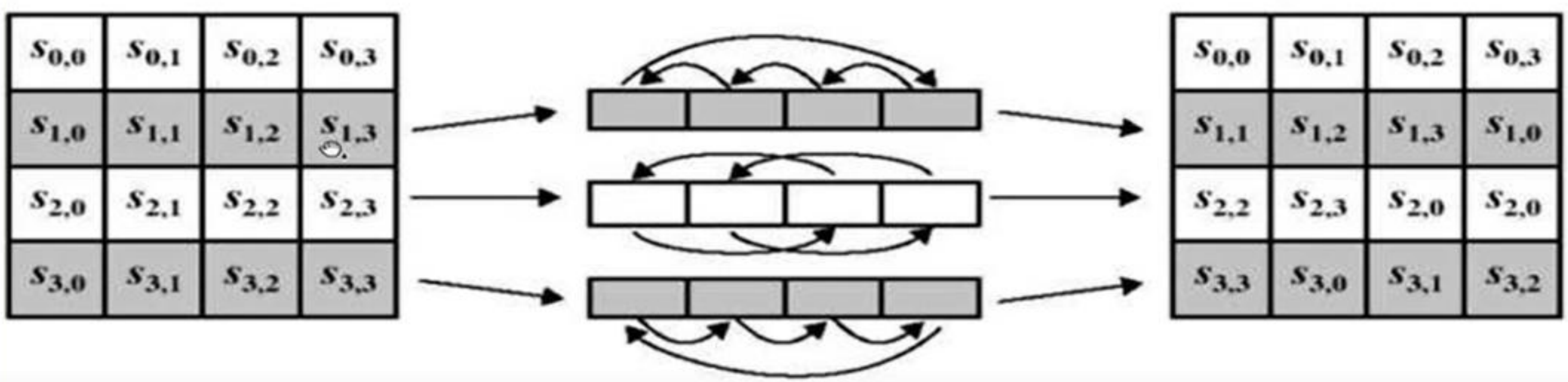
$$\begin{aligned} b_0 &= a_0 + a_4 + a_5 + a_6 + a_7 + c_0 \bmod 2 \\ &= 0 + 0 + 0 + 1 + 1 + 1 \bmod 2 \\ &= 1 \\ b_1 &= a_1 + a_5 + a_6 + a_7 + a_0 + c_1 \bmod 2 \\ &= 1 + 0 + 1 + 1 + 0 + 1 \bmod 2 \\ &= 0 \\ &\vdots \end{aligned}$$

- The result is: $(b_7 b_6 b_5 b_4 b_3 b_2 b_1 b_0) = (11101101)$, which is $\{ED\}$ in hexadecimal notation.

Algorithm MixColumn(c)
external: FieldMult, BinaryToField, FieldToBinary
for $i \leftarrow 0$ **to** 3
  do $t_i \leftarrow$ BinaryToField($s_{i,c}$)
$u_0 \leftarrow \text{FieldMult}(x, t_0) \oplus \text{FieldMult}(x+1, t_1) \oplus t_2 \oplus t_3$
$u_1 \leftarrow \text{FieldMult}(x, t_1) \oplus \text{FieldMult}(x+1, t_2) \oplus t_3 \oplus t_0$
$u_2 \leftarrow \text{FieldMult}(x, t_2) \oplus \text{FieldMult}(x+1, t_3) \oplus t_0 \oplus t_1$
$u_3 \leftarrow \text{FieldMult}(x, t_3) \oplus \text{FieldMult}(x+1, t_0) \oplus t_1 \oplus t_2$
for $i \leftarrow 0$ **to** 3
  do $s_{i,c} \leftarrow$ FieldToBinary(u_i)



AES key expansion.



ShiftRows stage.

Rcon Table

Round	Rcon	Round	Rcon
1	$(01000000)_{16}$	6	$(20000000)_{16}$
2	$(02000000)_{16}$	7	$(40000000)_{16}$
3	$(04000000)_{16}$	8	$(80000000)_{16}$
4	$(08000000)_{16}$	9	$(1B000000)_{16}$
5	$(10000000)_{16}$	10	$(36000000)_{16}$